

OncoVI without OncoKB: 7802 MTB variants

Vi Dang

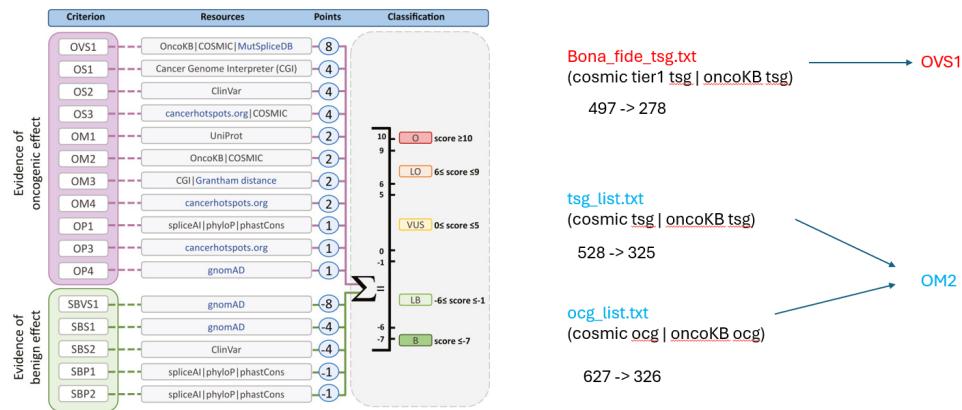
2026-08-11

Contents

1	Introduction	1
2	Results	2
2.1	7802 MTB variants	2
2.1.1	Confusion matrix	2
2.1.2	Variants with OncoVI score different between version 1 and version 2	3
2.1.3	Genes most frequently affected	3

1 Introduction

OncoVI (*Carta et al., Journal of Molecular Diagnostics, 2026*) integrates information from external databases subjected to licensing term such as COSMIC and OncoKB. OncoVI uses these resources to define the role of genes as tumor suppressor genes (TSGs) or oncogenes (OGs) for the evaluation of specific criteria. Here, we want to define the role of TSGs and OGs relying on COSMIC only and assess how OncoVI score and classification change between OncoVI version 1 (with OncoKB) and version 2 (without OncoKB). Data used for the comparison between the two versions: Molecular Tumour Board (MTB) set of 7,802 variants detected in more than 500 tumors analysed with the Illumina targeted gene panel TSO500 (analysed in OncoVI paper).



Adapted from Figure 1, *Carta et al., Journal of Molecular Diagnostics, 2026*.

Number of genes exclusive to OncoKB database and therefore missing in OncoVI version 2 (without OncoKB) compared to OncoVI version 1:

“bona_fide_tsg.txt”: 219

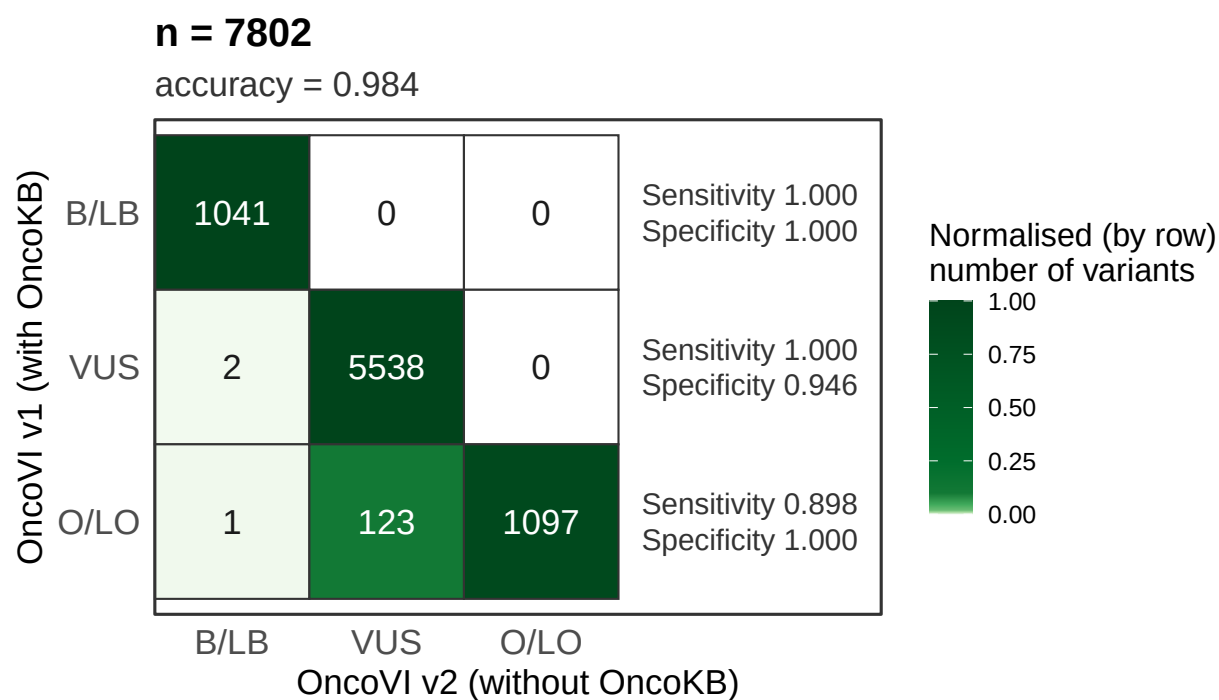
“tsg_list.txt”: 203

“ogs_list.txt”: 301

2 Results

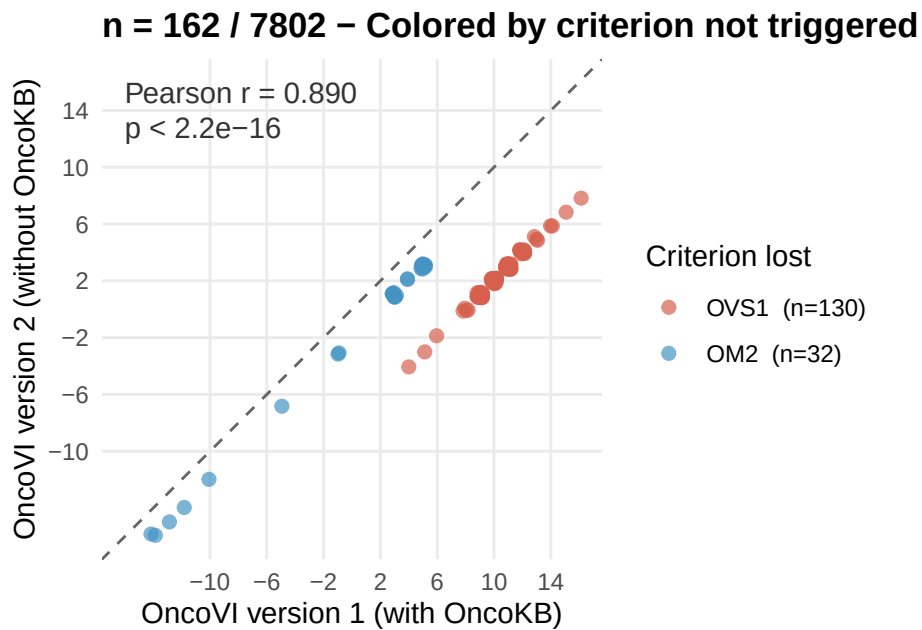
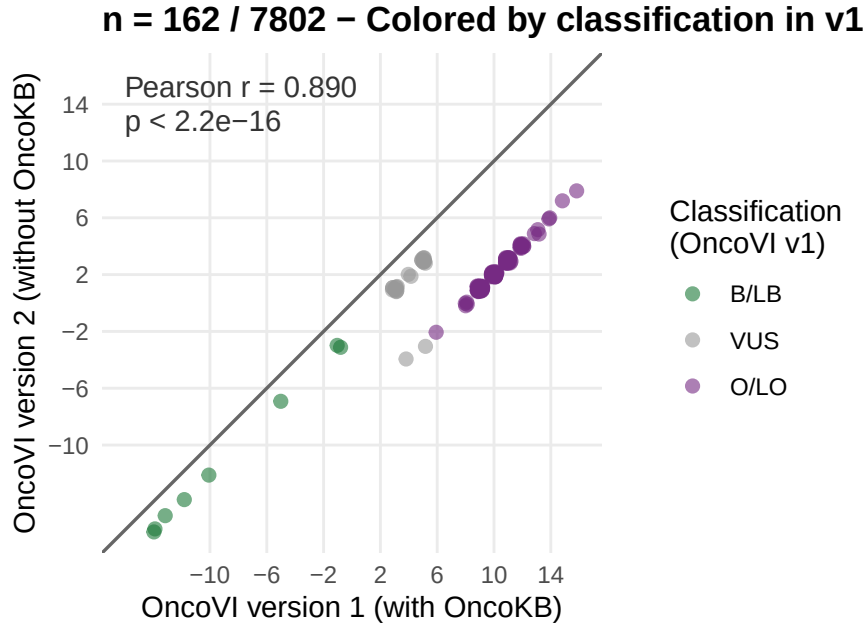
2.1 7802 MTB variants

2.1.1 Confusion matrix



Out of the total of 7802 variants, for 126 variants the classification changed, and for 162 variants the scores changed.

2.1.2 Variants with OncoVI score different between version 1 and version 2



When the score of a variant changes in v2, it is always lower than in v1, and either OVS1 or OM2 criterion is not triggered in v2.

2.1.3 Genes most frequently affected

The 162 variants with score changed between version 1 and version 2 belong to 59 different genes. Top affected genes:

Top 20 genes

